

HANGFUL

Requirements Specification Document

CMSI 4072: Senior Project II
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1. Introduction

This Requirements Specification Document identifies and describes all requirements for the Hangful application, including user stories, use cases, and acceptance criteria. It serves as the requirements traceability baseline for the CMSI 4072 Senior Project II course deliverable.

Hangful is a social utility iOS application that enables users to plan hangouts with friends, discover local deals at hangout venues, upload proof of attendance, and earn rewards. The application targets Gen Z users (18–28) in the Los Angeles area and is developed as a solo senior project using Swift/SwiftUI with mock data.

2. User Stories

2.1 First-Time User (Onboarded by Friend)

As a new user, I receive a Hangful link from a friend for a scheduled hangout at a specific place. I download Hangful, sign up with my phone number and OTP, and the scheduled hangout is automatically added to my feed. I can see the place, date, time, and which friends are going. This is the primary onboarding path — most users discover Hangful through a friend's shared hangout link rather than browsing the App Store.

2.2 User Wants to Hang Out

As a user, I want to create a hangout by tapping the + button or selecting a Quick Hangout category (Movie, Coffee, Matcha, Beach, Arcade). I name the hangout, pick a place from the directory (where I can see which places have active deals with reward badges), set the date and time, and confirm. The app generates a shareable link that I send to my friends via iMessage, WhatsApp, or any other app. Anyone who clicks the link gets the hangout added to their feed.

2.3 User Checks Who's Free

As a user, I want to open the Calendar tab to see which of my friends are available on a given day. I tap through the days of the week and see green bars under days where friends are free. For each day, I see a list of friends with their available time windows (e.g., '10:00am - 11:59pm') or their next available day if they're busy. This helps me pick the best day before creating a hangout.

2.4 User Finds Friends on the Map

As a user, I want to see where my friends are (by general area) and find good hangout spots near them. The Friend Map shows my friends as avatar pins on a MapKit map, with their last seen context below (e.g., '2 days ago (Matcha @ Santa Monica Pier)'). I can tap '+ Hangout Spot' to suggest a new location or 'Show Hangout Recs' to see recommended places.

2.5 User Earns a Reward

As a user who attended a hangout at a place with an active deal, I upload a proof photo showing I was there with friends. After admin approval, I receive an 8-character redemption code that I can copy and show at the business. My first proof approval also unlocks the referral feature, giving me a unique link to invite more friends to Hangful.

2.6 User Manages Friends

As a user, I want to manage my social circle on Hangful. I see my Best Friends (ranked by hangout frequency) in a horizontal scroll, my full friends list with last seen context, and my friend groups (like 'Horror Movie Squad'). I can view any friend's profile to see our shared hangout count, their bio, their availability, and our mutual friends. I can add new friends by search, phone number, or referral code.

3. Use Cases

3.1 UC-1: Authenticate via Phone OTP

Primary Actor: User. Precondition: User has the app installed. Main Flow: (1) User enters phone number, (2) system sends OTP, (3) user enters 6-digit code, (4) system verifies and authenticates. Postcondition: User is logged in and sees the Home tab. Alternate Flow: Invalid OTP displays error message; user can retry or go back to re-enter phone number.

3.2 UC-2: Create a Hangout

Primary Actor: Authenticated User. Precondition: User is logged in. Main Flow: (1) User taps + or Quick Hangout category, (2) enters hangout title, (3) selects a place from the directory (deal badges visible), (4) selects date and time, (5) reviews and confirms, (6) receives shareable link. Postcondition: Hangout appears in user's feed with 'confirmed' status. Alternate Flow: User cancels at any step; form resets.

3.3 UC-3: Submit Proof and Receive Code

Primary Actor: User who attended a hangout. Precondition: Hangout date has passed; hangout has a deal attached. Main Flow: (1) User opens hangout detail, (2) selects a photo from library, (3) previews and submits proof, (4) status changes to 'Pending Review', (5) admin approves, (6) redemption code is displayed. Postcondition: User has a copyable redemption code; referral feature is unlocked if this is their first approval.

3.4 UC-4: Check Friend Availability

Primary Actor: Authenticated User. Main Flow: (1) User opens Calendar tab, (2) taps a day of the week, (3) sees available friends with time windows and unavailable friends with next available day. Postcondition: User knows which friends are free and can create a hangout accordingly.

3.5 UC-5: View Friend Map

Primary Actor: Authenticated User. Main Flow: (1) User opens Friends tab, (2) taps map icon, (3) sees MapKit view with friend avatar pins, (4) sees friend list below with locations and last seen context. Postcondition: User has spatial awareness of where friends are located.

3.6 UC-6: Add a Friend

Primary Actor: Authenticated User. Main Flow: (1) User taps Add Friend, (2) selects method (search, phone, or referral code), (3) enters query, (4) sends friend request. Postcondition: Friend request is sent; recipient sees it in their requests banner.

4. Acceptance Criteria

ID	Feature	Acceptance Criteria
AC-001	Phone Auth	User can log in with any phone number using OTP code 123456 and lands on Home tab
AC-002	Hangout Creation	5-step flow completes successfully; hangout appears in feed with confirmed status and share link
AC-003	Quick Hangouts	Tapping a category pre-fills the title and opens the create flow at step 2
AC-004	Hangout Statuses	Cards display correct chips: confirmed (green), pending (yellow), cancelled (red)
AC-005	Place Deals	Places with deals show reward badges in the place picker; deal auto-attaches to hangout
AC-006	Proof Upload	User can select photo, preview it, submit, and see status change to Pending Review
AC-007	Redemption Code	After simulated approval, 8-char code displays with copy button
AC-008	Referral Unlock	Referral section on Profile shows locked state before first approval, unlocked after
AC-009	Friends List	Shows 9 seeded friends with avatars, names, locations, and last seen context
AC-010	Best Friends	Top 5 friends sorted by hangoutsTogether count displayed in horizontal scroll
AC-011	Friend Profile	Shows bio, hangout count, mutuals, action buttons, availability row
AC-012	Friend Groups	Groups tab shows 2 seeded groups with emoji, name, member count, stacked avatars
AC-013	Friend Requests	Pending request banner with Accept/Decline; accept adds to friends list
AC-014	Calendar	Day selector updates friend availability list; green bars show days with available friends
AC-015	Friend Map	MapKit displays friend avatar pins around LA with friend list below
AC-016	Explore	Place list filterable by category and deals-only toggle; place detail shows deal info

AC-017	Brand Request	Form submits successfully and shows confirmation message
AC-018	Logout	Confirmation dialog; returns to phone input screen

5. Requirements Traceability Matrix

The following matrix traces each functional requirement to the Swift file(s) that implement it and the acceptance criteria that validates it.

Req ID	Feature	Implementing File(s)	Test (AC)
FR-001-005	Authentication	AppAndAuth.swift, ViewModels.swift	AC-001
FR-010-019	Hangout Management	HangoutFeedView.swift, HangoutViews.swift	AC-002-005
FR-020-026	Proof & Redemption	HangoutViews.swift	AC-006-007
FR-030-040	Friends System	FriendsViews.swift	AC-009-013
FR-050-057	Calendar	CalMapProfile.swift	AC-014
FR-060-064	Friend Map	CalMapProfile.swift	AC-015
FR-070-074	Explore / Deals	ExploreView.swift	AC-016
FR-080-082	Referrals	CalMapProfile.swift (ProfileView)	AC-008
FR-090-092	Profile	CalMapProfile.swift	AC-017-018